Eric Hayden Campbell

Postdoctoral Research Fellow, UT Austin
Department of Computer Science
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Education

Cornell University
Ph.D. in Programming Languages, Dec 2024
Thesis Topic: Verified Configuration of Programmable Networks
2019 NSF GRFP Fellow

Pomona College
B.A.s in Computer Science & Mathematics, May 2017
Thesis: (In)finiteness and Linear Temporal Logic

Publications

PLDI '27 **Refractional Program Logic**
Eric Hayden Campbell, Mark Barbone, Eva Veciana, Neelu Kalani, and Nate Foster
In Submission

PLDI '27 **Robust Strategy Synthesis for Prediction-based Safety Games**
Eric Hayden Campbell, Aditya Akella, and Isil Dillig
In Submission

PLDI '27 **Reading Between the Lines: Finding Hidden Lemmas in Formal Proofs**
Maxine Xin, Eric Hayden Campbell, Sara Ansari, and Isil Dillig
In Submission

PLDI '26 **MatchBox: A Semantic Foundation for Data Plane Portability**
Eric Hayden Campbell, Robert Zhang, Divyanshu Saxena, Aditya Akella, and Isil Dillig
Distinguished Paper Award

PLDI '26 **Optimal Predicate Pushdown Synthesis**
Robert Zhang, Eric Hayden Campbell, Dixin Tang, and Isil Dillig

EuroSys '27 **Vulcan: Instance-Specialized, Verifiable Systems Heuristics Through LLM-Driven Search**
Rohit Dwivedula, Divyanshu Saxena, Sujay Yadalam, Eric Hayden Campbell, Daehyeok Kim, and Aditya Akella

OOPSLA '24 **Computing Precise Control Interface Specifications**
Eric Hayden Campbell, Hossein Hojjat, Nate Foster

NSDI '21 **Avenir: Managing Data Plane Heterogeneity via Control Plane Synthesis**
Eric Hayden Campbell, Bill Hallahan, Priya Srikumar, Carmelo Cascone, Jed Liu, Vignesh Ramamurthy, Hossein Hojjat, Robert Soulé, Ruzica Piskac, and Nate Foster

PLDI '22 **Kleene Algebra Modulo Theories**
Michael Greenberg, Ryan Beckett, and Eric Hayden Campbell
Distinguished Paper Award

POPL '22 Π4**: Dependently-Typed Data Plane Programming**
Matthias Eichholz, Eric Hayden Campbell, Matthias Krebs, Nate Foster, Mira Mezini

ECOOP '19 **How to Avoid Making a Billion-Dollar Mistake: Type-Safe Data Plane Programming with SafeP4**
Matthias Eichholz, Eric Hayden Campbell, Nate Foster, Guido Salvaneschi, Mira Mezini

CPP '23 **P4Cub: A Little Language for Big Routers**
Rudy Peterson, Eric Hayden Campbell, John Chen, Natalie Isak, Calvin Shyu, Ryan Doenges, Parisa Ataei, and Nate Foster

HotOS '25 **How I Learned to Stop Worrying and Love Learned OS Policies**
Divyanshu Saxena, Jiayi Chen, Sujay Yadalam, Yeonju Ro, Rohit Dwivedula, Eric Hayden Campbell, Aditya Akella, Christopher J. Rossbach, and Michael Swift

SCiL '21 **Epistemic Semantics in Guarded String Models**
Eric Hayden Campbell and Mats Rooth

TMS '16 **Constructing Integer Matrices with Integer Eigenvalues**
Christopher Towse and Eric Hayden Campbell
The Mathematical Scientist, June 2016

Industry

Fujitsu Laboratories of America
Research Intern, Property Inference for BGP Networks
Sunnyvale, CA. May 2018 - Aug 2018

Originate, Inc
Software Engineering Intern, Web Development
San Francisco, CA. May 2015 - Aug 2015

Gladly, Inc
Software Engineering Intern, Web Development
Palo Alto, CA. May 2014 - Aug 2014

Outreach

4H Career Explorations Conference
Workshop Designer & Leader, *Networking and Security*, 2018-2022

Expand Your Horizons Conference
Workshop Leader, *Googling With Paper Airplanes*, 2017-2019

Originate, Inc
Counselor, *Approachable Web Design*, 2015

Teaching

Academy of Math and Programming, Jane Street Capital
Advanced Instructor, CS3, Summer 2023, *Bespoke course design*
Instructor, CS3, Summer 2022, *Bespoke course design*

Computer Science Department, Cornell University
TA, OOP and Data Structures, Spring 2018, *Teaching Excellence Award*
TA, OOP and Data Structures, Fall 2017

Computer Science Department, Pomona College
TA, Databases Systems, Spring 2017
TA, Programming Languages, Fall 2016
Head TA, Introduction to Computer Science, Spring 2016
TA, Introduction to to Computer Science, Fall 2015